

Mathematics for Computing

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M.Tech in Computer Science and Engineering
(Artificial Intelligence and Software Engineering)
Semester I

6 August 2026

What is a Set?

Definition

A set is a well-defined collection of distinct objects called **elements**.

Mathematically,

$$A = \{a_1, a_2, a_3, \dots, a_n\}$$

where every element is unique.

Why are Sets Important in AI?

- Represent datasets
- Store detected objects
- Represent feature sets
- Query databases
- Build recommendation systems
- Define search spaces
- Knowledge representation

10 Examples in Set-Builder and Roster Form

No.	Set-Builder Form	Roster / Extended Form
1	$A = \{(-1)^{n+1} \cdot n^2 \mid n \in \mathbb{N}\}$	$\{1, -4, 9, -16, 25, -36, \dots\}$
2	$B = \left\{ \frac{n}{n+1} \mid n \in \mathbb{N}, 1 \leq n \leq 5 \right\}$	$\left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6} \right\}$
3	$C = \{x \in \mathbb{N} \mid 3 \mid x \wedge 5 \nmid x\}$	$\{3, 6, 9, 12, 18, 21, 24, \dots\}$
4	$D = \{x \in \mathbb{Z} \mid 1 \leq x \leq 3\}$	$\{-3, -2, -1, 1, 2, 3\}$
5	$E = \{w \in \{a, b\}^* \mid w \pmod{2} = 0\}$	$\{\varepsilon, aa, ab, ba, bb, aaaa, \dots\}$
6	$F = \{x \in \mathbb{Z}_{12}^+ \mid \gcd(x, 12) = 1\}$	$\{1, 5, 7, 11\}$ (Units of $\mathbb{Z}_{12}$)
7	$G = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x^2 + y^2 = 25\}$	$\{(0, \pm 5), (\pm 5, 0), (\pm 3, \pm 4), (\pm 4, \pm 3)\}$
8	$H = \left\{ \frac{1+(-1)^n}{2 \cdot n} \mid n \in \mathbb{N} \right\}$	$\left\{ 0, \frac{1}{2}, 0, \frac{1}{4}, 0, \frac{1}{6}, \dots \right\}$
9	$I = \{a^n b^n c^n \mid n \in \mathbb{Z}_{\geq 0}\}$	$\{\varepsilon, abc, aabbcc, aaabbbccc, \dots\}$
10	$J = \{w \in \{a, b\}^* \mid w = w^R\}$	$\{\varepsilon, a, b, aa, bb, aba, bab, \dots\}$

Cardinality of a Set

Definition

The cardinality of a set is the number of distinct elements in the set.

Mathematical Expression

$$|A| = n$$

where n is the number of elements in A .

AI Example

Suppose a YOLO object detection model detects the following objects in an image:

$$D = \{Car_1, Car_2, Person_1, TrafficLight_1, Bus_1\}$$

Therefore,

$$|D| = 5$$

Interpretation: The model detected **5 unique objects**.

Finite Set

Definition

A finite set is a set containing a limited (countable) number of elements.

Mathematical Expression

$$A = \{x \in \mathbb{N} \mid x \leq 5\} = \{1, 2, 3, 4, 5\}$$

$$|A| = 5$$

AI Practical Example

In a sentiment analysis model, each review is classified into one of three sentiment classes.

$$Sentiment = \{Positive, Neutral, Negative\}$$

$$|Sentiment| = 3$$

Since the number of sentiment classes is limited, **Sentiment** is a finite set.

Infinite Set

Definition

An infinite set contains an unlimited number of elements.

Mathematical Expression

$$A = \{x \in \mathbb{N}\} = \{1, 2, 3, \dots\}$$

AI Practical Example

An autonomous vehicle continuously generates LiDAR scans.

$$L = \{P_1, P_2, P_3, \dots\}$$

where P_i denotes the i^{th} LiDAR scan.

Since new scans are continuously generated,

L is treated as an infinite set.

Null (Empty) Set

Definition

A null (empty) set contains no elements.

Mathematical Expression

$$A = \{x \in \mathbb{N} \mid x < 0\} = \emptyset$$

Practical AI Example

An object detection model analyzes an empty parking lot.

Detected vehicles

$$DetectedCars = \emptyset$$

Cardinality

$$|DetectedCars| = 0$$

The model correctly detects no vehicles.

Equal Sets

Definition

Two sets are equal if they contain exactly the same elements.

$$A = B \iff \forall x, (x \in A \iff x \in B)$$

AI Example

$$GT = \{Car, Person, TrafficLight\}$$

$$P = \{TrafficLight, Person, Car\}$$

$$GT = P$$

Both sets contain identical labels.

Equivalent Sets

Definition

Two sets are equivalent if they contain the same number of elements.

$$|A| = |B|$$

Practical AI Example

Training Dataset

$$Cats = \{I_1, I_2, \dots, I_{5000}\}$$

Validation Dataset

$$Dogs = \{J_1, J_2, \dots, J_{5000}\}$$

Although images are different,

$$|Cats| = |Dogs| = 5000$$

Hence the datasets are equivalent.

Countable Set

Definition

A set is countable if its elements can be placed in one-to-one correspondence with the natural numbers.

$$A = \{a_1, a_2, a_3, \dots\}$$

Practical AI Example

Every image in a dataset receives a unique ID.

$$Images = \{img_1, img_2, img_3, \dots\}$$

Examples

- CIFAR-10
- COCO Dataset

Each image can be indexed uniquely.

Hence the dataset is countable.

Power Set

Definition

The power set of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{S \mid S \subseteq A\}$$

If $|A| = n$,

$$|\mathcal{P}(A)| = 2^n$$

AI Practical Example

Features used for training:

$$F = \{Age, Income, CreditScore\}$$

Possible feature subsets include

$$\emptyset, \{Age\}, \{Age, Income\}, \{Income, CreditScore\}, \dots, \{Age, Income, CreditScore\}$$

Since there are 3 features,

Subset

Definition

A set A is a subset of B if every element of A belongs to B .

$$A \subseteq B$$

AI Practical Example

All transactions:

$$T = \{T_1, T_2, T_3, T_4, T_5\}$$

Fraudulent transactions:

$$F = \{T_2, T_5\}$$

Therefore,

$$F \subseteq T$$

Interpretation: Every fraudulent transaction belongs to the set of all transactions.

Universal Set

Definition

The universal set contains all elements under consideration.

$$U = \{x \mid x \text{ belongs to the sample space}\}$$

Practical AI Example: Image Classification Dataset

Suppose a computer vision dataset contains images from five classes.

$$U = \{Cat, Dog, Car, Person, TrafficLight\}$$

A CNN classifier can predict only one of these classes.

Example subsets:

$$Animals = \{Cat, Dog\}$$

$$RoadObjects = \{Car, TrafficLight\}$$

Singleton Set

Definition

A singleton set contains exactly one element.

$$|A| = 1$$

Practical AI Example

A face recognition system identifies only one person in an image.

Prediction

$$\textit{Recognized} = \{ \textit{Tony} \}$$

Therefore

is a singleton set.

Union of Two Sets

Definition

The union contains every element belonging to either set.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Practical AI Example

Two object detection models analyze the same image.

YOLO detects

$$Y = \{Car, Person, Bus\}$$

Mask R-CNN detects

$$M = \{Person, Bike, TrafficLight\}$$

Combined detections

$$Y \cup M = \{Car, Person, Bus, Bike, TrafficLight\}$$

Intersection of Two Sets

Definition

The intersection of two sets contains the elements common to both sets.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

AI Practical Example

Movies liked by User A:

$$A = \{M_1, M_2, M_3, M_5\}$$

Movies liked by User B:

$$B = \{M_2, M_3, M_4\}$$

$$A \cap B = \{M_2, M_3\}$$

Interpretation: These common movies can improve collaborative recommendations.

Set Difference

Definition

The difference of two sets contains elements in the first set but not in the second.

$$A - B = \{x \mid x \in A, x \notin B\}$$

AI Practical Example

$$M = \{M_1, M_2, M_3, M_4, M_5\}$$

$$W = \{M_2, M_4\}$$

$$M - W = \{M_1, M_3, M_5\}$$

Interpretation: Recommend movies not yet watched.

Symmetric Difference

Definition

The symmetric difference contains elements that belong to exactly one of the two sets.

$$A \triangle B = (A - B) \cup (B - A)$$

AI Practical Example

Features selected by LASSO:

$$L = \{Age, Income, CreditScore\}$$

Features selected by Decision Tree:

$$D = \{Income, Education, Experience\}$$

$$L \triangle D = \{Age, CreditScore, Education, Experience\}$$

Interpretation: These features are selected by only one of the two algorithms.

Complement

Definition

The complement of a set contains every element of the universal set except those in the given set.

$$A^c = U - A$$

Practical AI Example

Universal Set

$$U = \{All\ Emails\}$$

Spam Emails

$$S = \{Spam\}$$

Complement

$$S^c = NonSpam$$

Cartesian Product

Definition

The Cartesian product is the set of all ordered pairs.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

AI Practical Example

Images:

$$I = \{Image_1, Image_2\}$$

Languages:

$$L = \{English, Spanish\}$$

$$I \times L = \{(Image_1, English), (Image_1, Spanish), (Image_2, English), (Image_2, Spanish)\}$$

Interpretation: Each pair represents generating a caption for an image in a specific language.

General Form of Inclusion–Exclusion Principle

Definition

The Inclusion–Exclusion Principle computes the total number of distinct elements present in the union of several overlapping sets.

For two sets,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

General Formula

$$\left| \bigcup_{i=1}^n A_i \right| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|$$

Relation

Definition

A relation R from A to B is a set of ordered pairs where elements of A are related to elements of B .

$$R \subseteq A \times B$$

Data Science Example: Customer–Product Relation Customers:

$$C = \{C_1, C_2, C_3\}$$

Products:

$$P = \{P_1, P_2, P_3\}$$

Suppose customers purchased products:

$$R = \{(C_1, P_1), (C_1, P_3), (C_2, P_2), (C_3, P_1)\}$$

$$R \subseteq C \times P$$

Interpretation: Each pair represents a customer–product purchase and can be used for recommendation systems.

Equivalence Relation

Definition

A relation R is an **equivalence relation** if it is

Reflexive, Symmetric, and Transitive

Data Science Example: Customer Records Let

$$D = \{d_1, d_2, d_3, d_4\}$$

Define

$$d_i R d_j \iff \text{CustomerID}(d_i) = \text{CustomerID}(d_j)$$

Suppose

$$ID(d_1) = ID(d_3) = 101$$

$$ID(d_2) = ID(d_4) = 102$$

Then

$$[d_1] = \{d_1, d_3\}, \quad [d_2] = \{d_2, d_4\}$$

Interpretation: Records with the same Customer ID belong to the same equivalence class.

Partially Ordered Set (Poset)

Definition

A relation R on a set A is a **partial order** if it is

- Reflexive
- Antisymmetric
- Transitive

Then (A, R) is called a **Partially Ordered Set (Poset)**.

Example

Let

$$A = \{1, 2, 3, 4, 5\}, \quad aRb \iff a \leq b$$

Examples:

$$1 \leq 2, 2 \leq 4, 3 \leq 5, 1 \leq 5$$

Since every pair of elements is comparable under $\leq$, this relation is a partial order (in fact, a total order). Hence, (A, R) is a **Poset**.

Hasse Diagram of a Poset

Let

$$A = \{1, 2, 3, 4, 5\}, \quad aRb \iff a \leq b$$



$$1 \prec 2 \prec 3 \prec 4 \prec 5$$

This Hasse diagram forms a chain (total order).

Power Set Ordered by Subset Relation

Consider

$$A = \{1, 2, 3\}$$

Its power set is

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Define the relation

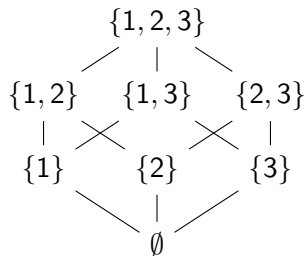
$$S_i R S_j \iff S_i \subseteq S_j$$

$$(P(A), \subseteq)$$

forms a **Partially Ordered Set (Poset)**.

Hasse Diagram of the Power Set

$$A = \{1, 2, 3\}, \quad R = \subseteq$$



Hasse diagram of the power set ordered by the subset relation.

Chain

Definition

A subset C of a Poset is called a **chain** if every pair of elements in C is comparable.

$$\forall a, b \in C, \quad a \leq b \text{ or } b \leq a$$

Example

Let

$$A = \{1, 2, 3, 4, 5\}, \quad aRb \iff a \leq b$$

Consider

$$C = \{1, 3, 5\}$$

Since

$$1 < 3 < 5$$

every pair of elements is comparable.

C is a Chain

Antichain

Definition

A subset S of a Poset is called an **antichain** if no two distinct elements of S are comparable.

$$a \neq b \Rightarrow a \not\leq b \text{ and } b \not\leq a$$

Example

Let

$$A = \mathcal{P}(\{1, 2, 3\})$$

ordered by the subset relation $\subseteq$. Consider

$$S = \{\{1\}, \{2\}, \{3\}\}$$

Since

$$\{1\} \not\subseteq \{2\}, \quad \{2\} \not\subseteq \{3\}, \quad \{1\} \not\subseteq \{3\}$$

no two elements are comparable.

S is an Antichain

Thank You!